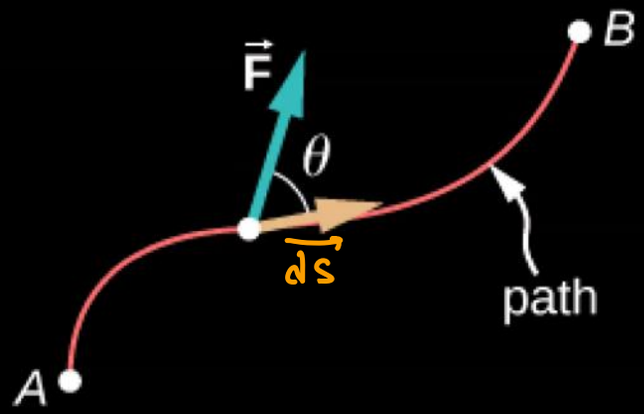
Work: If a Force $\vec{F}$ acts on a particle which gets displaced through $d\vec{s}$ then elemental work done by the force for that displacement is defined as $dW = \vec{F} \cdot d\vec{s}$



→ Here $d\vec{s}$ is displacement of the particle on which the force is acting. (And not displacement of the "body").

→ If $\vec{F} \perp d\vec{s}$ then work done is zero.

→ $\vec{F} \cdot d\vec{s} = F\,ds\cos\theta = F\,ds_{\parallel}$ where $ds_{\parallel}$ is the component of displacement parallel to the force.

→ $\vec{F} \cdot d\vec{s} = F\cos\theta\,ds = F_{\parallel}\,ds$ where $F_{\parallel}$ is component of the force // to $d\vec{s}$.

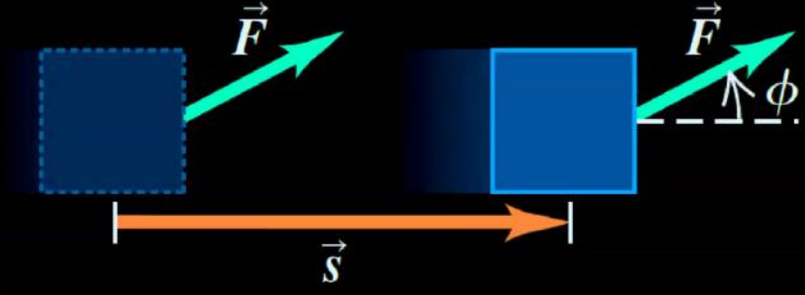
→ For finite displacements work can be found by integrating the elemental work, i.e:

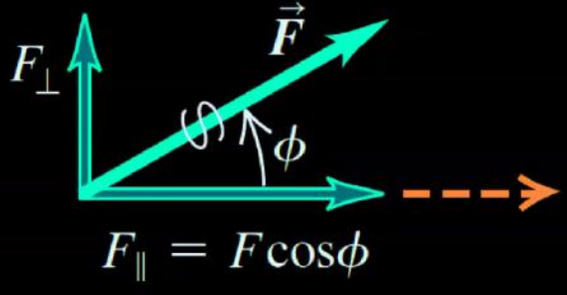
$$\Delta W = \int \vec{F} \cdot d\vec{s}$$

Work Done By A Constant Force:

If a force is constant (both magnitude and direction), the expression for work becomes simply:

$$\Delta W = \vec{F} \cdot \vec{S} = F S_{\parallel} = F_{\parallel} S$$





This can be shown as follows: $dW = \vec{F} \cdot d\vec{s} \Rightarrow \Delta W = \int \vec{F} \cdot d\vec{s}$

however if $\vec{F}$ is constant, we can take out $\vec{F}$.

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outside the integral (∵ "·" is a distributive operation). Thus:

$$dW = \vec{F} \cdot \int d\vec{s} = \vec{F} \cdot \vec{s} = Fs_{\parallel} = F_{\parallel} s.$$

→ If angle between $\vec{F}$ and $\vec{s}$ is obtuse, work done is negative, if acute work done is positive and no work is done if $\vec{F} \perp \vec{s}$.

→ Work done is frame dependent (since displacement is frame dependent).

EXAMPLE: Suppose you are petting a dog (stroking from head towards tail) while the dog is standing perfectly still,

(a) From ground frame, what is the sign of work done by you on the dog and the sign of work done by dog on you

(b) From your hand's frame, what is the sign of work done by you on the dog and sign of the work done by the dog on you. (Treat Dog like a rigid body)

Solution: Always remember to look at displacement of the particle on which force is acting (to figure out the sign of work done). (a) From ground frame, particles forming the Dog's surface are not moving at all. So work done on Dog is zero, while particle of the hand are moving with frictional force acting against the motion of the hand. Thus work done by the dog on your hand is negative.

(b) From hand's frame, Dog's particles are moving forward while friction on them is acting backwards, thus work done on dog is negative, while particle of the hand are at rest in this frame ∴ work done on hand is zero from hand's frame.

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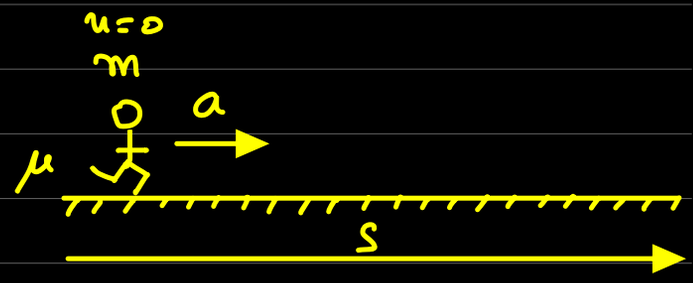
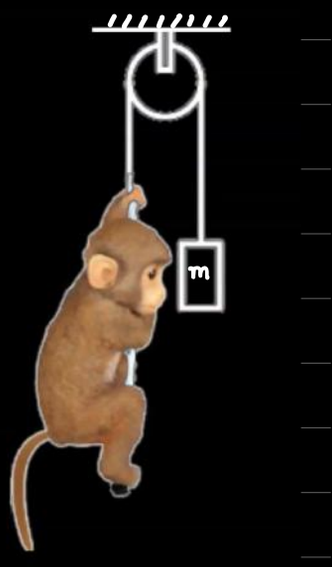
EXAMPLE: A man of mass m starts running (without slipping) with a constant acceleration a. By the time he covers a distance S, how much work is done by the frictional force on the man?



Solution: Since the man runs without slipping, there is no displacement of particles of his shoes while friction is acting on them (one foot is fixed on the ground and the other foot is moving), while the particles of the foot which is moving are not experiencing any frictional force. Thus net work done by frictional force is zero.

EXAMPLE: A monkey hangs in equilibrium being counterbalanced by a mirror of equal mass. If now monkey climbs up a distance h relative to the rope. From the ground frame, (a) Is the work done by the rope on the monkey positive negative or zero (b) What is the work done by the tension force on the monkey assuming that monkey climbed very slowly. (Initially the system is at rest, monkey's hands do not slip on the rope).



Solution: Imagine the climbing process very carefully. Since forces on the mirror and monkey are identical, their motions are also identical, thus as the monkey climbs up, the block moves up by the same distance and the rope must be passing to the left side over the pulley. Thus monkey's hand holding the rope must be going down by exactly the same distance (since rope is not slipping relative to the hand). Thus if block goes up a distance y, the monkey's hand goes down by a distance y while the tension acting on it is mg

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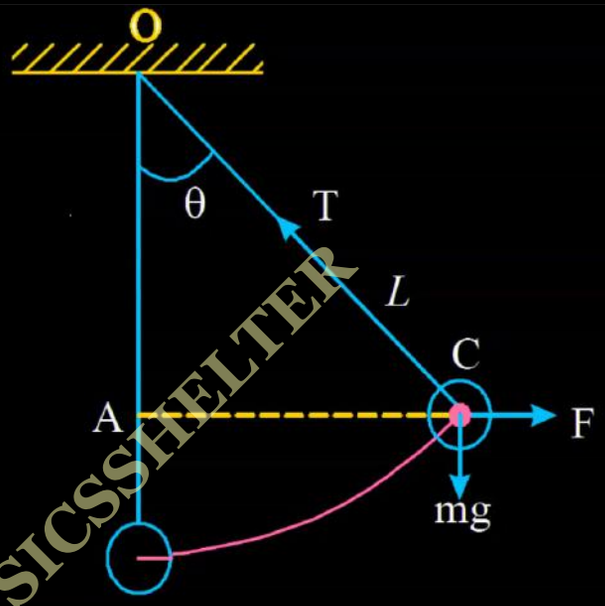
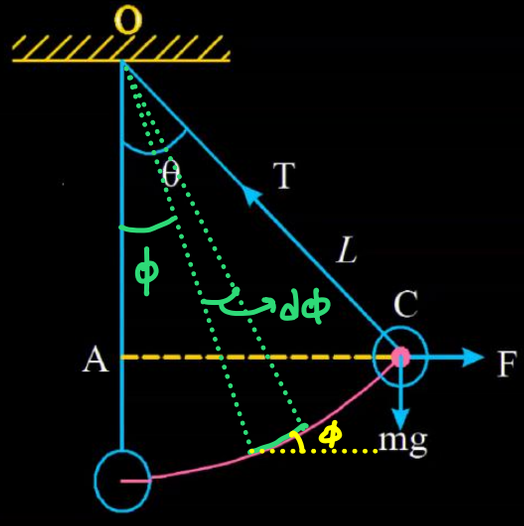
upwards and work done in this process on the monkey is $dW = -mgy$. Now in this process, relative to the rope, monkey's head has risen a distance $2y$. Thus, $2y = h$ and the required work is then $dW = -mgy = -mg\frac{h}{2}$ Ans.

EXAMPLE: A bob hangs in equilibrium from a string of length L. Now it is slowly pulled to an angular position θ (see figure) by a gradually changing horizontal force. Find (a) Work done by the horizontal force (b) Work done by gravitational force mg.



Solution: Consider an elemental displacement through an angle $d\phi$ at a general angular position ϕ. Here, by Lami's theorem:

$$\frac{F}{\sin(180^\circ-\phi)} = \frac{mg}{\sin(90^\circ+\phi)} \Rightarrow F = mg\tan\phi$$



Thus $dW = mg\tan\phi \times L\,d\phi \times \cos\phi$

$$\Rightarrow dW = mgL\sin\phi\,d\phi$$

$$\Rightarrow W = mgL\int_0^\theta \sin\phi\,d\phi = mgL(1-\cos\theta)$$ Ans (a).

(b) Now mg is a constant force $\therefore$ we can simply find its work using $W = FS_{\parallel}$, $S_{\parallel} = -L(1-\cos\theta)$ ($\because S_{\parallel}$ is against mg i.e anti parallel). Thus work done by mg:
$W = -mgL(1-\cos\theta)$ Ans

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Work Energy Theorem In 1 Dimension: Suppose a force F_x acts on a body moving along x axis with initial velocity u_x and final velocity v_x then we can write:

$$F_x = m v_x \frac{dv_x}{dx} \Rightarrow \int_{u_x}^{v_x} F_x dx = m \int_{u_x}^{v_x} v_x dv_x$$

i.e $\Delta W = \frac{1}{2} m (v_x^2 - u_x^2)$, thus area under the $F-x$ graph represents change in kinetic energy (Here particle is moving only along x axis and Force is always along x axis).

Work Energy Theorem (General): A small displacement in 3D space can be written as $d\vec{s} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$. Now let a force acting on a particle be $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$, thus elemental work: $dW = \vec{F}\cdot d\vec{s} = F_x dx + F_y dy + F_z dz$

Now $F_x = m v_x \frac{dv_x}{dx}$ similarly $F_y = m v_y \frac{dv_y}{dy}$ and $F_z = m v_z \frac{dv_z}{dz}$

Thus $\Delta W = \int F_x dx + F_y dy + F_z dz = m\left(\int_{u_x}^{v_x} v_x dv_x + \int_{u_y}^{v_y} v_y dv_y + \int_{u_z}^{v_z} v_z dv_z\right)$

Thus $\Delta W = \frac{m}{2}\left[v_x^2 - u_x^2 + v_y^2 - u_y^2 + v_z^2 - u_z^2\right]$

$\Rightarrow \Delta W = \frac{m}{2}\left[(v_x^2 + v_y^2 + v_z^2) - (u_x^2 + u_y^2 + u_z^2)\right]$

$= \frac{m}{2}\left[v^2 - u^2\right]$. Thus we can formally state work energy theorem as: **Net work done by a force on a particle is equal to change in its kinetic energy.**

Work Energy Theorem For Systems: Since a system is composed of several particles and work energy theorem is applicable for each particle, we can as well write the work energy theorem for systems:

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→ Net work done by all the forces acting on a system (both internal as well as external) is equal to change in kinetic energy of the system.

→ Internal force can do work. (Think of a bomb blast, or a person initially still starts shaking / moving his limbs).

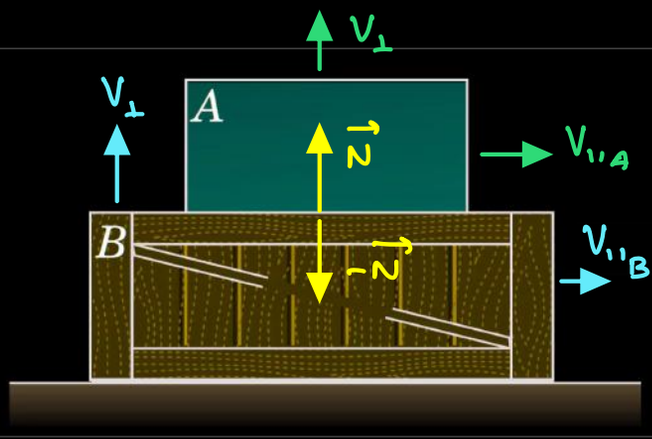
→ Sum of work done by internal force in a rigid body is zero because at every points (particle interfaces), forces are action reaction pairs and displacements are equal, thus work done by action and reaction have opposite signs but same magnitudes.

POWER: Rate of work done by a force is called instantaneous power, Mathematically, $P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{s}}{dt}$ but $d\vec{s}/dt$ is velocity thus:

$$\boxed{P = \vec{F} \cdot \vec{v}}$$

ZERO WORK PAIRS: Following pairs of forces do no net work when considered as action reaction pairs:

1. NORMAL REACTION PAIR: Whenever two surfaces are in contact, their component of velocity along the normal is same (i.e velocity of approach of particles towards each other is zero). Also normal reactions on two surfaces are equal and opposite. Now power developed by Normal Pair on the surfaces, $P = \vec{N} \cdot \vec{v}_A + (-\vec{N} \cdot \vec{v}_B) = Nv_1 - Nv_1 = 0$ (Recall that $\vec{a} \cdot \vec{b} = ab_{\parallel}$). Now since rate of work done by the pair is zero therefore total work done by the pair is also zero.



QED.

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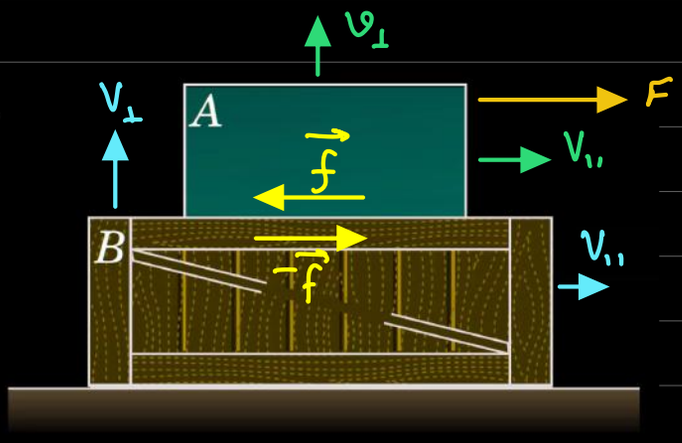
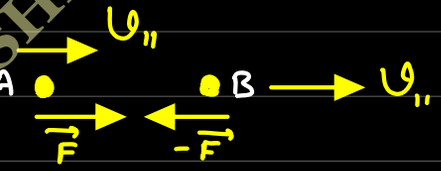
2. Static Friction Pair:



Whenever there exists static friction between a pair of surfaces, the velocity along the contacting surfaces is same for particles of both bodies while frictional forces are equal and opposite, thus power of the static friction pair, $P = \vec{f}\cdot\vec{v}_A + -\vec{f}\cdot\vec{v}_B$

$= -f v_{||} + f v_{||} = 0$ QED

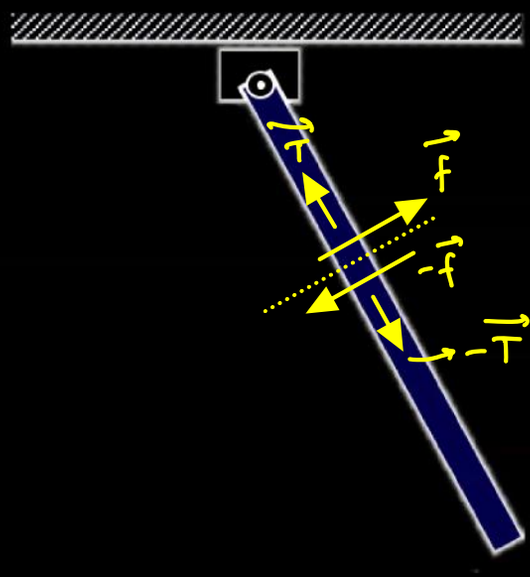
3. Internal Force Pairs In Inextensible Strings:



In an inextensible string, velocity of approach or separation is zero between adjacent particles, thus their in component of velocity along the string is same (say $v_{||}$).

However the forces of interaction between adjacent particles are action reaction pairs and act along line joining them. Thus power developed by the pair: $F v_{||} - F v_{||} = 0$ thus pair work of internal forces in inextensible strings is zero (for extensible strings this can be non zero since particles could be moving with different velocities along the string).

4. Internal Force Pairs In Rigid Bodies:



In any rigid body, the separation between any two particles remains constant.

We can mentally draw a surface in a rigid body and imagine it to be interface between two parts of the rigid body. The shearing component of interaction force may be likened to static friction pair (which has zero work as a pair) similarly the normal component of interaction may be likened to normal

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reaction pair (which also has zero work as a pair). Thus internal force pairs in rigid bodies have zero net work (as a pair).

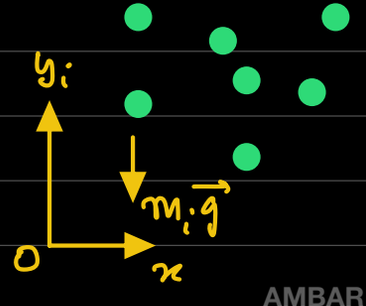
Proof Of Tension Work Method For Constraints: Whenever we have a system of massless strings, frictionless and massless pulleys and blocks the net work done by tension on all massive objects is zero. This can be explained from the following step by step argument.

1. Since strings and pulleys are massless, they cannot possess any kinetic Energy, ∴ net work done on them must be zero
2. If we combinedly consider the system of massless pulleys and massless, smooth, inextensible strings, excluding massive objects then work done by all internal forces of the system is zero (∵ pulleys are rigid bodies, strings are inextensible ∴ internal force pair works cancel, and strings interact with massless pulleys through normal reaction pairs, whose work is also zero).
3. Therefore work done by massive objects on string must also be zero, (∵ massless systems can't have any K.E.)
4. Whenever string connect with massive objects, work done by strings on massive objects has same magnitude but opposite sign as work done by strings on massive objects. ∴ Work done by strings on massive objects must also be zero. QED.

(This is because contacting particles of strings with blocks have same velocities but opposite force directions being action reaction pairs).

Work Done By Gravity On A System Of Particles:

Consider a system of particles as shown in the figure. Let m_i be the mass of i^{th} particle having its Y coordinate as y_i, Suppose this particle's height changes



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by an amount Δy_i then work done on this particle must be $(-m_i g \hat{j}) \cdot (\Delta y_i \hat{j}) = -m_i g \Delta y_i$, thus total work done on the system of particles must be:

$$\Delta W = -g \sum m_i \Delta y_i = -g \left[(\Sigma m_i y_i)_{final} - (\Sigma m_i y_i)_{initial} \right]$$

Now let M be the total mass of the system i.e Σm_i. then we can also write the work done as:

$$\Delta W = -Mg \left[\left(\frac{\Sigma m_i y_i}{M} \right)_{final} - \left(\frac{\Sigma m_i y_i}{M} \right)_{initial} \right]$$

The quantity $\frac{\Sigma m_i y_i}{M}$ is called height of centre of mass of the particles. For simple bodies like a uniform rod, cuboid, cylinder or a sphere, this corresponds to height of the geometrical centroid. For other general shaped, you will learn locating centre of mass in the next chapter). Thus gravitational work can be simply written as:

$$\Delta W_g = -Mg (h_{final} - h_{initial}) = -Mg \Delta h$$

Where h represents height of centre of mass of the body with some horizontal reference line.

→ For bodies in pure translation (not changing orientation), every particle undergoes same displacement, therefore Δh can be found just by looking at any particle of the rigid body however for rotating bodies, or non rigid bodies (or shape changing) bodies, we must locate height of CM only to find the work done by gravity).

→ Even though height of CM will depend on the chosen reference, the change in height will be independent of that so for finding work done we can choose the reference suitably. (Usually we choose a reference in which either initial or final height of CM is zero).

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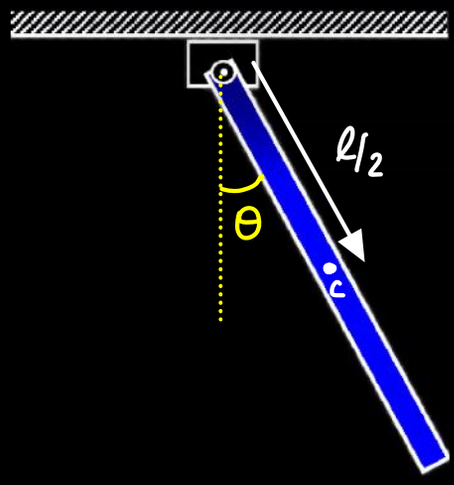
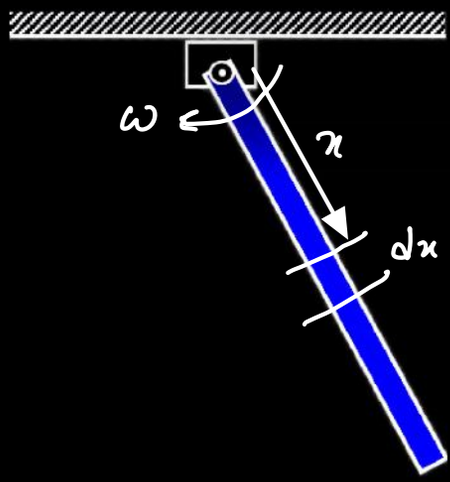
EXAMPLE: A thin rod of mass m and length l begins to fall from a horizontal position while pivoting about a frictionless pivot as shown. Find the angular speed of the rod when it is making an angle θ with vertical.



Solution: If we choose the pivot level as the reference, initial height of CM = 0 and final height of CM $= -\frac{l}{2}\cos\theta$. Thus work done by gravity in the process is $\Delta W = -mg\Delta h = mg\frac{l}{2}\cos\theta$. This must be the change in kinetic energy of the rod.

Now consider an element of the rod having a thickness dx. If ω is the angular speed of the rod then elemental kinetic energy:

$$dKE = \frac{1}{2}\left(\frac{m}{l}dx\right)(\omega x)^2, \text{ thus Net KE:}$$



$$KE = \frac{m\omega^2}{2l}\int_0^l x^2 dx = \frac{ml^2\omega^2}{6}$$

Thus by work energy theorem:

$$mg\frac{l}{2}\cos\theta = \frac{ml^2\omega^2}{6} \Rightarrow \omega = \sqrt{\frac{3g\cos\theta}{l}} \quad \text{Ans.}$$

→ Sometime centre of mass of an entire object may not be known directly, however we might know CM of its different pieces. In such cases, we can find the work done piece by piece.

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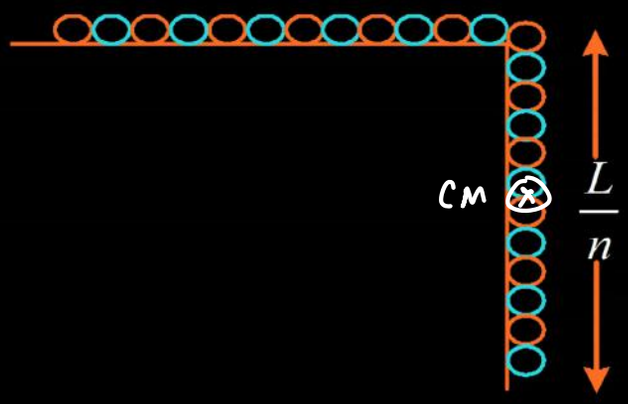
EXAMPLE: A uniform chain of mass M and length L is lying on a smooth table with 1/n fraction hanging. Find the work required by an external agent to completely pull the chain up. (Slowly)

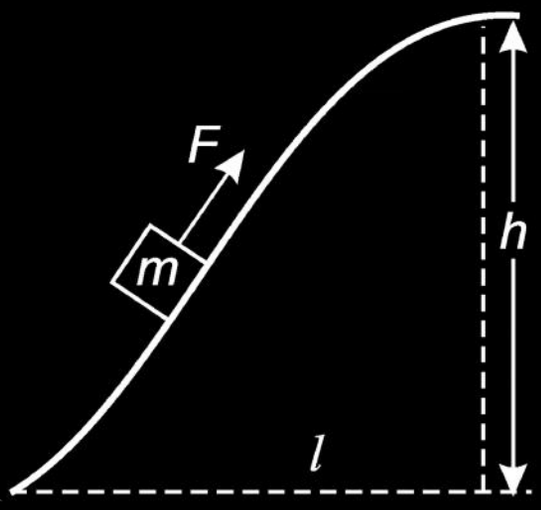


Solution: By work Energy theorem, sum of work done by gravity and external agent must be zero.
Here we can just consider the work done by gravity on the hanging part (because the top portion doesn't change height). Now work done by gravity:

$$W_g = -\frac{m}{n} g \left[0 - \left(\frac{-L}{2n}\right)\right] \qquad (\because W_g = -mg\Delta L).$$

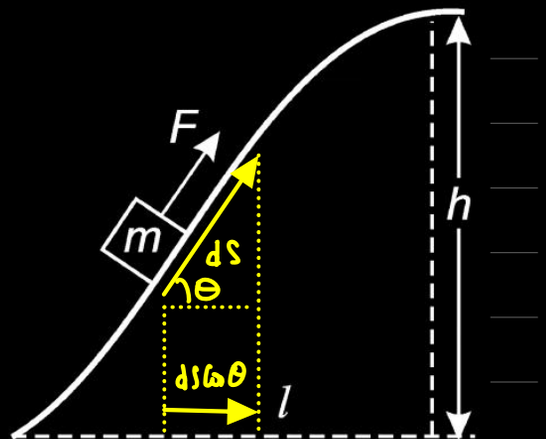
$$= -\frac{mgL}{2n^2}, \quad \text{Now } W_{ext} + W_g = 0 \Rightarrow W_{ext} = -W_g = \frac{mgL}{2n^2}$$

TROOV 1.121: A body of mass m was slowly hauled up a hill by a force F always directed along tangent to the trajectory. Find the work done by this force if height of the hill is h, and coefficient of friction is μ.



Solution: Slowly hauling means that work done by all forces is zero in total.
Thus $W_F + W_g + W_f = 0 \Rightarrow$

$W_F = -W_g - W_f$. Now let us find work done by friction first. Consider an elemental displacement ds. Here work done by friction is $dW_f = -\mu mg\cos\theta\, ds$. Now recognize that $ds\cos\theta$ is shadow of the displacement on the horizontal line. Thus $dW_f = -\mu mg\, ds_{shadow}$,



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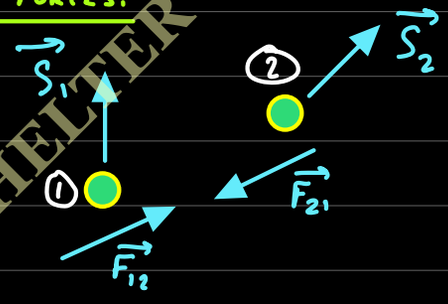
Now integrating, $W_f = -\mu mg \int ds_{slider} = -\mu mgl$

$W_G = -mgh$, thus $W_F = -W_f - W_G = \mu mgl + mgh$ Ans.

→ Recognize that here exact shape of the hill did not matter. If the motion were not slow then we could not write $N = mg\cos\theta$ and this technique would not work.

Frame Independence Of Net Work Done By Internal Forces:



Consider a system of two particles. Let $\vec{F}_{12}$ be the force on ① due to ② and $\vec{F}_{21}$ be the force on ② due to ①, further let S_1 and S_2 be their displacements. Then in the process of displacement, work done by their interaction force pair:

$$W_{12} = \vec{F}_{12}\cdot\vec{S}_1 + \vec{F}_{21}\cdot\vec{S}_2 = \vec{F}_{12}\cdot(\vec{r}_1 - \vec{r}_{o1}) - \vec{F}_{12}\cdot(\vec{r}_2 - \vec{r}_{o2})$$

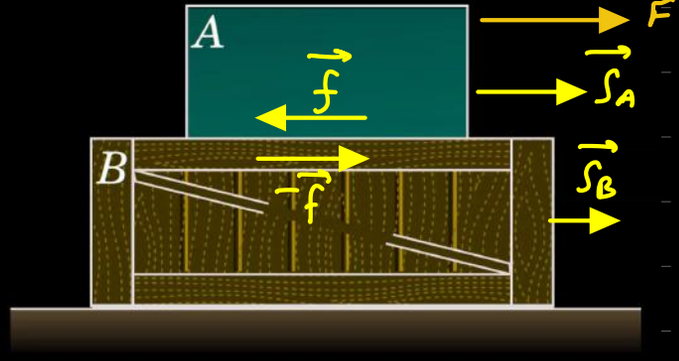
where $\vec{r}_{o1}$ and $\vec{r}_{o2}$ are initial positions, also $\vec{F}_{21} = -\vec{F}_{12}$ (:: 3rd law).

Thus:

$W_{12} = \vec{F}_{12}\cdot(\vec{r}_{12} - \vec{r}_{o12})$, Now $\vec{r}_{12}$ is relative position of ① as seen from ②, which is independent of frame, similarly $\vec{r}_{o12}$ is independent of frame, thus work done by the internal force pair between ① & ② is frame independent. Now if we have system of large number of particles, we can always see the force interaction amongst particles as internal action reaction pairs. Thus, net work done by all the internal forces in a system is independent of frame.

Work Done By Kinetic Friction Pair:

Consider a system of two blocks as shown in the figure. Here Kinetic friction is an internal force pair between two blocks. Thus work done by kinetic friction pair must also be frame independent.



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Thus if f is the kinetic frictional force, the work done by the pair is simply $\vec{f}\cdot\vec{s}_A - \vec{f}\cdot\vec{s}_B = \vec{f}\cdot(\vec{s}_A - \vec{s}_B) = -f(s_{AB})$ where

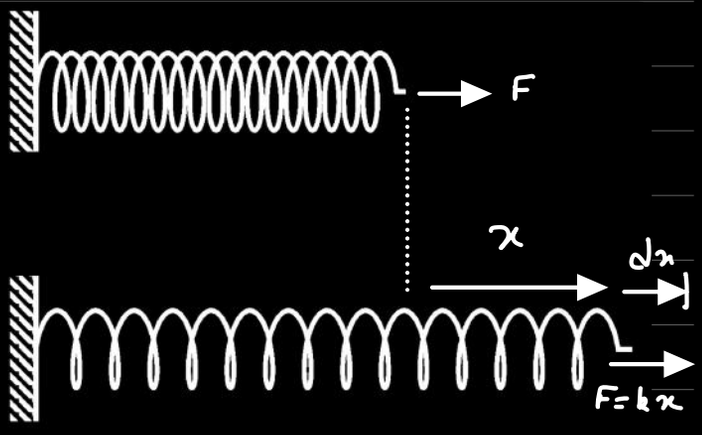
s_{AB} represents the distance through which one block slips on the other. (Note that work done by kinetic friction pair is always negative, we can always go in rest frame of one of the surfaces and see it directly. In such a frame only one block moves on which the friction acts against the displacement). Thus W_f:

$$\boxed{W_f = -f(s_{AB})}$$ s_{AB}: Relative slipped distance.

→ If friction force is variable, pair work can be found by integrating from frame of one of the surfaces.

Work Done By Internal Forces In A Massless Spring:

Consider a spring being stretched by applying a force. When it is stretched by a force, through a distance x, applied force must be kx (∵ massless object must have net force zero, considering FBD of last particle of the spring). Now if it is further stretched by a distance dx, the elemental work done by external force: $dW = kx\,dx$ thus, total work done by external force in stretching the spring, :



$$W = \int_0^x kx\,dx = \frac{1}{2}kx^2.$$ Now since massless spring cannot have any kinetic energy therefore sum of work done by internal forces and external forces must be zero. Thus work done by internal forces in spring.

$$\boxed{W_s = -\frac{1}{2}kx^2}$$ → Here x could either be compression or elongation from natural length.

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FORCE FIELD: If the force on a particle placed somewhere in space depends only on its position (and not velocity/acceleration etc), the particle is said to be subjected to a force field. Examples of force fields could be gravitational field and electrostatic field. Force fields could be classified either as conservative (work done in a displacement is independent of path) or non-conservative (work done in a displacement depends on path).

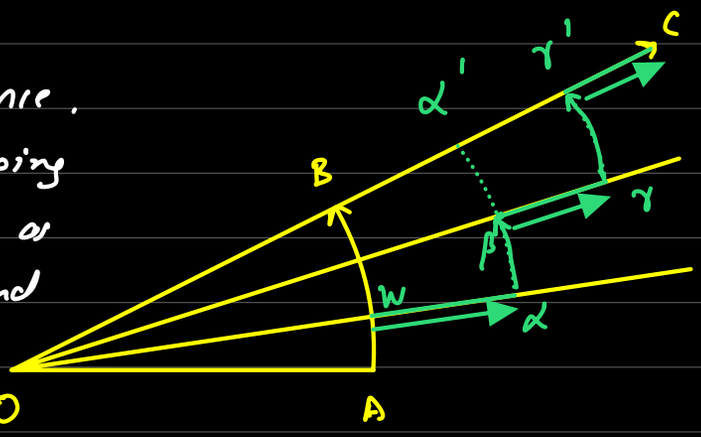
CENTRAL FORCES AND FORCE FIELDS: A force field which points radially and whose magnitude only depends on distance from a point (and not on angular position) is called a central force field. Mathematically central force has the form:

$$\vec{F} = F(r)\,\hat{r}$$ where $F(r)$ is any arbitrary function of r.

Example of central forces could be the electric field due to a point charge or the gravitational field due to a point mass.

PATH INDEPENDENCE OF WORK DONE IN CENTRAL FORCES:

Let O be the centre of some central force. We will show that work done in going from O to C along OABC is same as that along OAWαβγγ'C and extend the argument to arbitrary paths. Here the paths considered have only radial and tangential segments.



Because the forces are central, the work done along any arc centred at O must be zero and work done for any radial displacement must be same regardless of the the angular position. Now, due to radial symmetry, work done along W → α is same as that along B → α' similarly work done along β → γ is same as that from α' → γ', thus work done along OABC is same as that along OAαβγγ'C. However our second path is completely arbitrary, we could as well having infinite number of

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radial and tangential segments in our second path and still work done will be same. Thus for all central forces, work done is independent of path. (This also means that work done in a closed path must be zero because one obvious path along which work done is zero is the path of zero length i.e no movement!)

Path Independence Of Work In Superposition Of Central Forces:

Let O_1, O_2, O_3 etc be the centres of central fields with corresponding force fields $\vec{F_1}(r)$, $\vec{F_2}(r)$, $\vec{F_3}(r)$ etc then as shown in the previous section,

$\int \vec{F_1}\cdot d\vec{s}$, $\int \vec{F_2}\cdot d\vec{s}$ and $\int \vec{F_3}\cdot d\vec{s}$ etc are all independent of path. Now if we have all the three central forces acting simultaneously then net work done:

$$W = \int (\vec{F_1}+\vec{F_2}+\vec{F_3})\cdot d\vec{s} = \int \vec{F_1}\cdot d\vec{s} + \int \vec{F_2}\cdot d\vec{s} + \int \vec{F_3}\cdot d\vec{s}$$

(∵ dot product is distributive). Now each term on RHS is path independent thus LHS must also be path independent.

→ Thus superposition of central forces leads to field which have the property of path independence of work

→ Uniform fields can also be thought of as central fields with force centres at infinity.

Conservative Forces: The forces for which work done is independent of path are known as conservative forces (Alternatively, Forces for which work done in a closed path is zero are known as conservative forces). Thus all central forces and their superpositions are conservative forces.

→ Since work done by conservative forces in a closed path is zero, we can also say that work done against conservative forces is retrievable. For instance when a ball is thrown up with some KE, it loses its KE as it gains height due to negative work done by gravity, however, it regains all of it back once it

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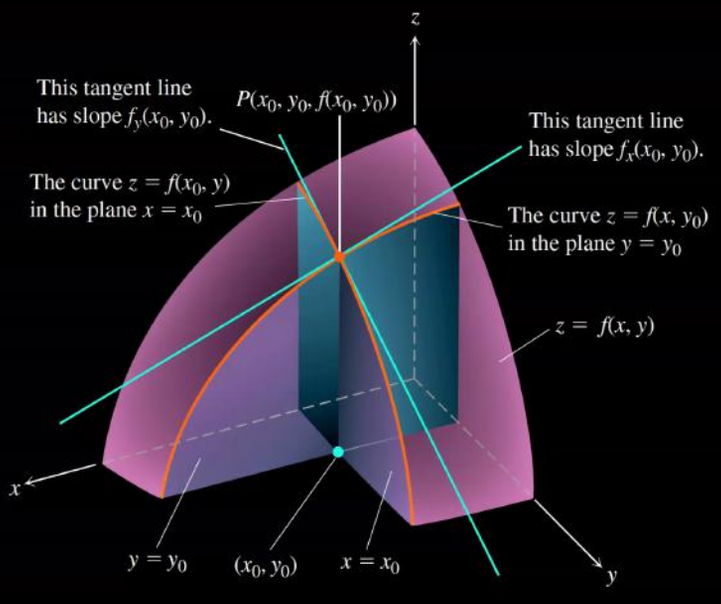
returns due to an equal amount of positive work in the returning process. Thus conservative forces do not cause any loss of energy to the surroundings and hence they are also called non-dissipative forces. Examples of conservative forces are gravity, electrostatic forces and spring forces.

NON-CONSERVATIVE FORCES: The forces for which work done depends on path are called non conservative forces (alternatively, forces for which work done in closed path is not identically zero are non-conservative forces). Examples of non conservative forces are, kinetic friction, viscous drag and non-electrostatic electric forces (will be covered in class XII in chapter on electromagnetic induction). Non conservative forces are also called dissipative forces because when they work, the energy is dissipated in to the surroundings in the form of heat, light and/or sound. This lost energy is not retrievable even if the body is restored back to its original position (∵ work done in closed path is non-zero).

PARTIAL DERIVATIVES:

Imagine a hill, whose height z is given as a function of the floor coordinates (x, y) such that $z = f(x, y)$. Now imagine a point $(x_0, y_0, f(x_0, y_0))$ on surface of the hill. If someone asks, "what is the slope of the hill at this point", then that question would be ambiguous because the hill will have different slopes in different directions. So we will have to talk about directional slopes. We could for example talk about slope along y direction or slope along x direction. Such slopes are called directional slopes (or directional derivatives). If we move a knife ∥ to yz plane to cut the surface, we will get a planar curve whose x coordinate will be constant. Slope of such



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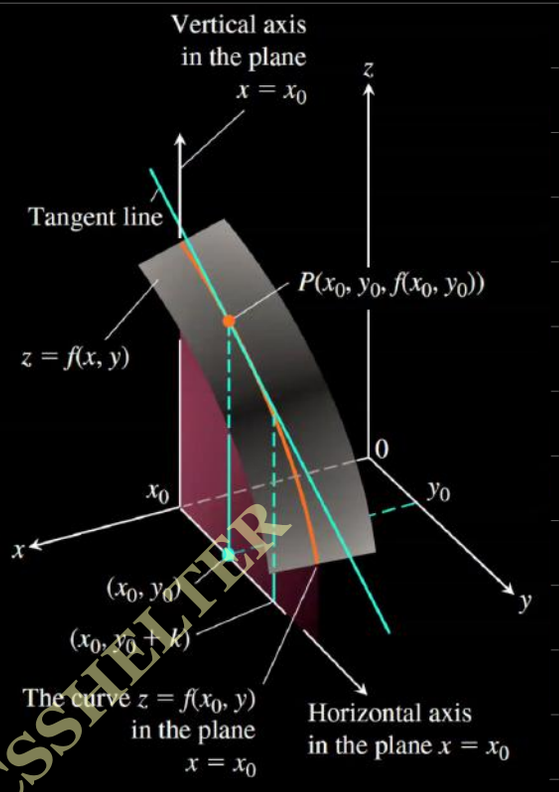
a curve is called y directional derivative (or partial derivative wrt y coordinate). Thus we mathematically define partial derivative wrt y at a general point $z = f(x, y)$ as:

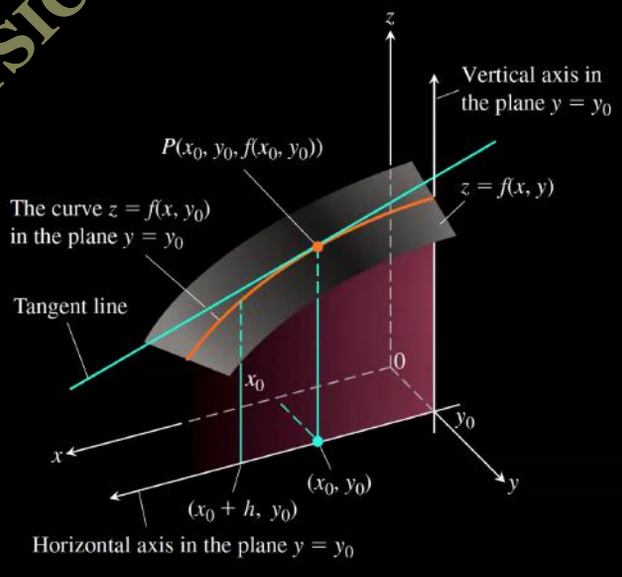
$$f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = \lim_{\Delta y \to 0} \left[\frac{f(x, y+\Delta y) - f(x, y)}{\Delta y} \right]$$

(Note that we kept x constant). This is the slope of the red curve (planar) in the adjacent figure). The tangent at this point is shown in the green colour. We can define the partial derivative wrt x direction in a similar manner. Thus, mathematically:



$$f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = \lim_{\Delta x \to 0} \left[\frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} \right]$$

(Note that here we kept y as constant). This represents slope of the surface in direction parallel to x-axis which is also the slope of the red curve in adjacent figure (with green line being the tangent defining directional slope).



→ The idea of partial derivatives can also be extended to multivariable functions of any number of variables. While calculating partial derivative wrt any particular direction, all other variables are treated as constant. For instance if Temperature (T) were a function of space coordinates (x, y, z) then we could define partial derivative of temperature in z direction as:

$$\partial_z T = \frac{\partial T}{\partial z} = \lim_{\Delta z \to 0} \left[\frac{T(x, y, z+\Delta z) - T(x, y, z)}{\Delta z} \right]$$

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EXAMPLE: If $f(x,y) = x^2 + 3xy + y - 1$, find $\partial f/\partial x$ and $\partial f/\partial y$ at point $(4,-5)$.

Solution: For finding $\partial f/\partial x$ we treat y as constant thus:

$$\frac{\partial f}{\partial x} = 2x + 3y \quad @(4,-5) \quad \frac{\partial f}{\partial x} = 8 - 15 = -7. \text{ Ans.}$$

Now for finding $\partial f/\partial y$ we treat x as constant. Thus, $\partial f/\partial y$:

$$\frac{\partial f}{\partial y} = 3x + 1 \quad @(4,-5), \quad \frac{\partial f}{\partial y} = 12 + 1 = 13 \text{ Ans.}$$

EXAMPLE: Find f_x and f_y if $f(x,y) = 2y/(y+\cos x) = 2y(y+\cos x)^{-1}$

Solution: $f_x = -2y(y+\cos(x))^{-2} \times (-\sin(x))$ ∴ (chain rule)

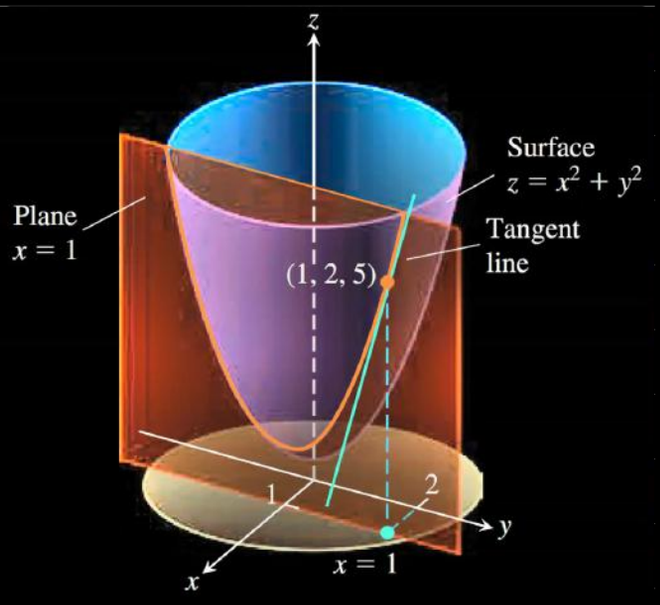
thus $f_x = 2y\sin(x)/(y+\cos x)^2$. Ans.

Now $f_y = 2\left[\dfrac{(y+\cos x) - y(1)}{(y+\cos x)^2}\right] = \dfrac{2\cos x}{(y+\cos(x))^2}$ Ans.

EXAMPLE: The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$ in a parabola. Find the slope of tangent to the parabola at $(1,2,5)$ (see figure).

Solution: slope $= \dfrac{\partial z}{\partial y} = 2y = 4$ Ans.

(Alternatively if $x = 1$, parabola is simply $z = y^2 + 1$ thus $\partial z/\partial y = 2y = 4$).



TANGENT PLANE APPROXIMATION:

For sufficiently small Δx, Δy and Δz, $f(x+\Delta x, y+\Delta y, z+\Delta z)$ can be approximated as: $f(x+\Delta x, y+\Delta y, z+\Delta z) = f(x,y,z) + \dfrac{\partial f}{\partial x}\Delta x + \dfrac{\partial f}{\partial y}\Delta y + \dfrac{\partial f}{\partial z}\Delta z$.

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$$f(x+\Delta x, y+\Delta y, z+\Delta z) \approx f(x,y,z) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \frac{\partial f}{\partial z}\Delta z$$

Second Order Partial Derivatives: Just like ordinary derivatives we can also have higher order partial derivatives like $\partial^2 f/\partial x^2$ or f_{xx} $\partial^2 f/\partial y^2 : f_{yy}$, $\partial^2 f/\partial x \partial y = f_{xy}$ and $\partial^2 f/\partial y \partial x$ or f_{yx}.

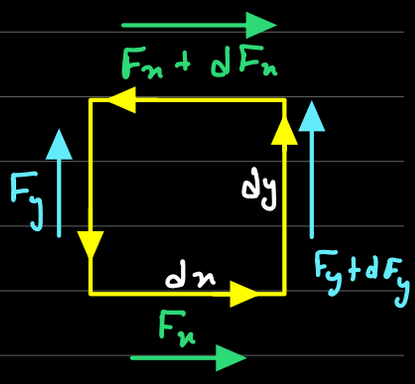
$$\rightarrow \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$ [order of partial derivative does not matter].

[This is called mixed derivative theorem and holds whenever f_x, f_y, f_{xy} and f_{yx} are defined in the region around a point (a,b) and are continuous at a,b].

Test Of Conservative Force Fields (Cartesian):

Let a force field be defined as:

$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$ where F_x, F_y and F_z are some functions of x, y and z. Now consider an infinitesimal rectangular loop in the xy plane having sides dx and dy. While travelling around this loop, net work done must be zero if the force were conservative. Thus for conservative force:



$$F_x dx + (F_y + dF_y)dy - (F_x + dF_x)dx - F_y dy = 0$$

$$\Rightarrow \quad -dF_x dx + dF_y dy = 0$$ Now dividing by $dx\,dy$:

$$\frac{dF_y}{dx} - \frac{dF_x}{dy} = 0 \quad \text{or} \quad \frac{dF_x}{dy} = \frac{dF_y}{dx}$$, however since F_x and F_y are multivariable functions therefore by derivative, here we mean partial derivatives, thus for conservative forces:

$$\boxed{\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}}$$ Similarly $$\boxed{\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}}$$ and $$\boxed{\frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}}$$

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(∵ we could build similar argument by considering the loops in yz or zx planes).

THE NABLA '$\vec{\nabla}$' OPERATOR: For sake of compactness of notation, while writing equations in multivariable calculus and vector calculus, it is often convenient to write in terms of the '∇' operator which is defined as

$$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

→ For vectors the operator may be used as a dot or a cross, for instance if we have a vector $\vec{F}$ we could do an operation either $\vec{\nabla} \times \vec{F}$ (known as curl of $\vec{F}$) or $\vec{\nabla} \cdot \vec{F}$ (known as divergence of $\vec{F}$). (Will be illustrated shortly)

→ For scalar fields this operator is directly applied. For instance if $U(x,y,z)$ is some scalar field we could find $\vec{\nabla} U$.

TEST OF A CONSERVATIVE FIELD IN TERMS OF CURL:

If $\vec{F}$ is a conservative force field, then its curl is zero i.e for conservative force fields $\vec{\nabla} \times \vec{F} = 0$ (identically). This can be shown as follows: Let $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$ then $\vec{\nabla} \times \vec{F}$ is calculated using matrix method similar to cross product.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix} = \hat{i}\left[\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right] - \hat{j}\left[\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\right] + \hat{k}\left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right]$$

Now we showed in a previous section that for conservative force fields, $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$ and $\frac{\partial F_z}{\partial x} = \frac{\partial F_x}{\partial z}$ and $\frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}$ thus the matrix expansion must be identically zero for conservative fields. QED.

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→ For conservative force fields $\boxed{\vec{\nabla} \times \vec{F} = 0}$ on we can say that curl of conservative fields is zero.

→ This test must be used very carefully. It is valid only if $\vec{F}$ is continuous and differentiable everywhere in the region inside the considered loop, otherwise this test may fail.

EXAMPLE: Consider the force field $\vec{F} = y^2\hat{i} + z^2\hat{j} + x\hat{k}$. Is the field conservative?

Solution: $F_x = y^2$, $F_y = z^2$, $F_z = x$ $\quad \frac{\partial F_y}{\partial x} = 0$, $\frac{\partial F_x}{\partial y} = 2y$

$\because \partial F_y/\partial x \neq \partial F_x/\partial y \quad \therefore$ field is non-conservative.

A SPECIAL FIELD: ZERO CURL BUT STILL NON CONSERVATIVE:

Consider the force field $\vec{F} = \frac{-y}{x^2+y^2}\hat{i} + \frac{x}{x^2+y^2}\hat{j} + 0\hat{k}$

Here $\frac{\partial F_x}{\partial y} = \frac{(x^2+y^2)\times(-1) - (-y)(2y)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$$\frac{\partial F_y}{\partial x} = \frac{(x^2+y^2)\cdot 1 - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

thus we see that $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$ so the field appears to be conservative however the field is not defined at the origin so the test fails for any path that encloses the origin i.e the field behaves like a conservative field so long as origin is not inside the path. (or work done in closed path is zero if the path doesn't encircle the origin). We will get more insight in to this field by expressing it in polar coordinates. Realize that $x^2+y^2 = r^2$, Also,

$\hat{\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j} = -\frac{y}{\sqrt{x^2+y^2}}\hat{i} + \frac{x}{\sqrt{x^2+y^2}}\hat{j} = \frac{1}{r}(-y\hat{i} + x\hat{j})$

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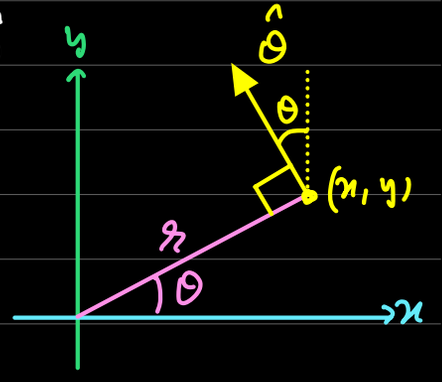
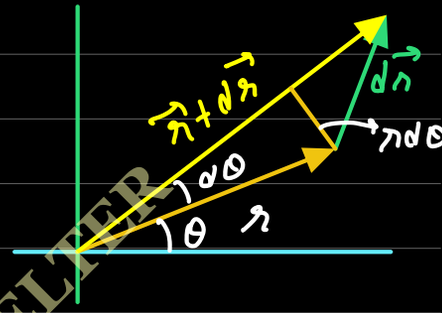
Thus our force field is simply. $\vec{F} = \frac{1}{r^2} \cdot r\hat{\theta} = \frac{1}{r}\hat{\theta}$



Now any elemental displacement in the 2D plane can be written as $d\vec{r} = r\,d\theta\,\hat{\theta} + dr\,\hat{r}$, thus elemental work for a small displacement is given as:

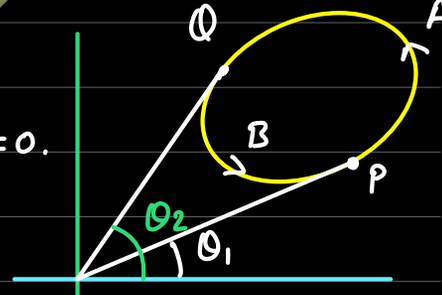
$$dW = \vec{F} \cdot d\vec{r} = \frac{1}{r}\hat{\theta} \cdot (r\,d\theta)\hat{\theta}$$



i.e $dW = d\theta$.

Now consider an arbitrary path PAQBP not enclosing the origin. Work done along this path is simply $W_{PAQ} + W_{QBP}$

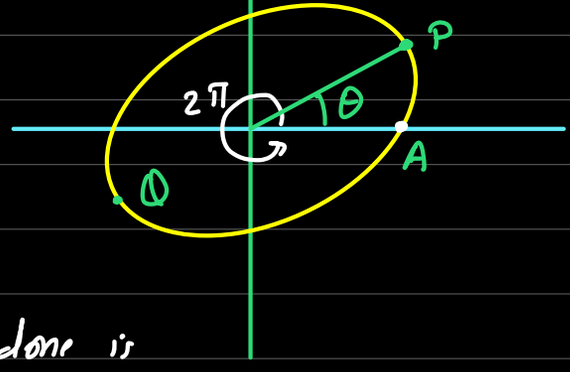
$$= \int_{\theta_1}^{\theta_2} d\theta + \int_{\theta_2}^{\theta_1} d\theta = (\theta_2 - \theta_1) + (\theta_1 - \theta_2) = 0.$$



thus the work done for any closed path not enclosing the origin is zero.

Now consider a path going around the origin. (Path APQA), for this path,

$$\int dW = \int_0^{2\pi} d\theta = 2\pi \neq 0 \quad \text{thus}$$



for path going around the origin work done is non-zero. Therefore we classify this force as non-conservative.

→ Infact if we make n laps around the origin, work done will be $2n\pi$.

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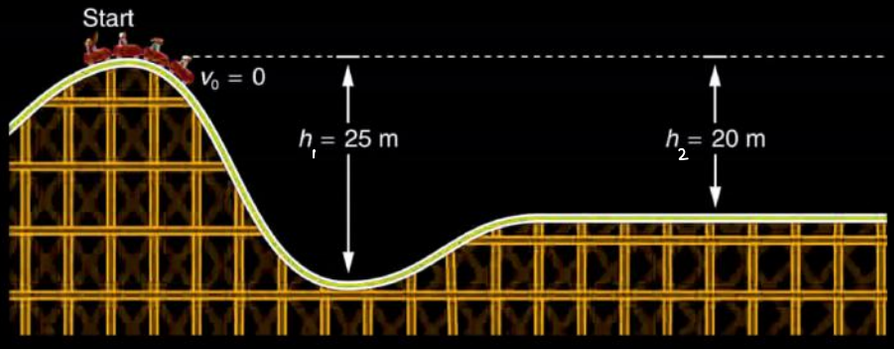
Change In Potential Energy: (ΔU) : Negative of work done by a conservative force is known as change in potential energy. Suppose we move from a point A in space to another point B under action of conservative force $\vec{F}_c$ then the potential energy change is defined as:

$$\Delta U = \int_{\vec{r}_A}^{\vec{r}_B} -\vec{F}_c \cdot d\vec{r} = U_B - U_A$$

Note that the change in potential energy does not depend upon the path followed in going from A to B because of conservative nature of the force. As an example, suppose a block is lifted up through a height h then change in work done by gravity is $-mgh$ and therefore, change in potential energy is $\Delta U = -W_g = +mgh$.

Potential Energy: Negative of work done in bringing a system to a given configuration from certain arbitrarily fixed reference configuration is called potential energy. The choice of reference can be completely arbitrary. This can also be called the change in potential energy of a system in bringing it from certain reference configuration to the given configuration. Potential Energy is never absolute due to arbitrary choice of reference, however change in potential energy is absolute.



For instance consider a toy train of mass m starting from top of a hill. If the deepest point of the valley is considered as reference config. initial potential energy is mgh_1 i.e 25 mg while if plateau is considered as reference then its PE is mgh_2 i.e 20 mg. Now suppose train reaches lower point of the valley, the by first reference, final PE is 0 and change in PE, $\Delta U = 0 - 25mg = -25mg$. Now if plateau were chosen as reference then at deepest point of the

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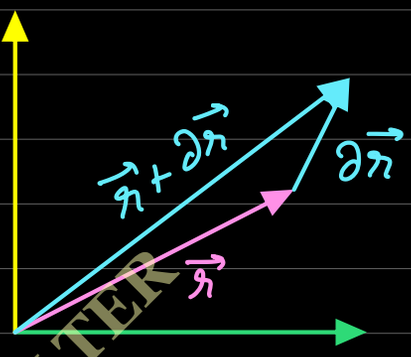
valley, $U = mg \times (-5) = -5mg$ thus $\Delta U = -5mg - 20mg = -25mg$.
thus we get same value of ΔU even though U values are different depending on the reference levels.

Finding Conservative Force Function From Potential Energy Function:

Let $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$ be some conservative force and let $U(x,y,z)$ be the associated potential energy function wrt certain reference. Let $\partial\vec{r}$ be an infinitesimal displacement of a particle from $\vec{r}$ to $\vec{r} + \partial\vec{r}$ then:



$\partial\vec{r} = \partial x\,\hat{i} + \partial y\,\hat{j} + \partial z\,\hat{k}$ where ∂x, ∂y and ∂z are increments in x, y & z coordinates.
Thus work done by conservative force in the process:

$$dW = \vec{F}\cdot\partial\vec{r} = -dU \Rightarrow dU = -\vec{F}\cdot\partial\vec{r} = -(F_x\partial x + F_y\partial y + F_z\partial z)$$

Now if we were to move $\parallel$ to x axis keeping y & z coordinates constant, $\partial y = \partial z = 0$ then:

$$\partial U = -F_x\partial x \text{ thus:}$$

$$\boxed{F_x = -\frac{\partial U}{\partial x}} \quad \text{similarly} \quad \boxed{F_y = -\frac{\partial U}{\partial y}} \quad \text{and} \quad \boxed{F_z = -\frac{\partial U}{\partial z}}$$

Combining all the three, $\boxed{\vec{F} = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]}$

Now recall that we defined the nabla '$\vec{\nabla}$' operator as:
$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ thus using this notation we can compactly write the Force function in terms of potential energy function as:

$$\boxed{\vec{F} = -\vec{\nabla}U}$$ when $\vec{\nabla}$ is thus operated on a scalar field, the operation is called gradient thus, we sometimes also say $\vec{F} = -\text{Grad}(U)$ or the conservative force is negative of the gradient of potential energy.

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EXAMPLE: Potential Energy in a region of space is given as $U = x^2 + xy + z$ Joules (x, y & z are in metres). (a) Find the force function (b) Find the force on the particle at point (2,2,2). (c) Find work done by external agent in moving the particle from (1,1,1) to (2,2,2) very slowly.

Solution: $\vec{F} = -\vec{\nabla}U = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$ thus:

(a) $\vec{F} = -\left[(2x+y)\hat{i} + x\hat{j} + 1\hat{k}\right]$ Ans

(b) Now $\vec{F}$ @ (2,2,2) $= -\left[6\hat{i} + 2\hat{j} + \hat{k}\right]$ Ans.

(c) By work energy theorem, work done by conservative force and the external agent must add up to zero $\because \Delta KE = 0$, thus

$W_{ext} + W_c = 0 \Rightarrow W_{ext} = -W_c = \Delta U$

thus, $W_{ext} = U(2,2,2) - U(1,1,1) = 10 - 3 = 7$ Joule Ans.

MECHANICAL ENERGY: The summation of kinetic energy and potential energy of any system is called its mechanical energy. Mathematically, $ME = U + KE$.

COROLLARIES OF WORK ENERGY THEOREM:

1. **CHANGE IN MECHANICAL ENERGY:** Net work done by forces other than the conservative forces on a system is equal to change in its mechanical energy. (This includes work done by forces of unknown origin also). [Acronym: CHIME]

PROOF: According to work energy theorem, net work done by all the forces acting on a system is equal to change in the kinetic energy of the system. Thus:

$$W_c + W_o = \Delta KE \quad (1)$$

(c: conservative, o: other)

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Now rearranging equation ①:

$$W_0 = \Delta KE - W_c$$

$$\Rightarrow \quad W_0 = \Delta KE + \Delta U = \Delta ME \quad \text{QED.}$$

$$\boxed{\Delta W_0 = \Delta ME}$$

2. **Conservation Of Mechanical Energy: (COME):** If in any system, work done by other forces is zero then mechanical energy of the system remains constant.

Proof: Using CHIME, $\Delta W_0 = \Delta ME$ thus if $\Delta W_0 = 0 \Rightarrow \Delta ME = 0$ thus $ME =$ constant.

→ This can be alternatively stated in following ways:

① $KE_i + PE_i = KE_f + PE_f$ (i: initial f: final).

or initial mechanical energy is equal to final mechanical energy.

Alternatively:

② $PE_i - PE_f = KE_f - KE_i$ or "loss in PE = Gain in KE"

(this statement is often easier to apply because we don't need any references for changes in PE).

or ③

$KE_i - KE_f = PE_f - PE_i$ or "loss in KE = Gain in PE".

→ For most situations we like to use ② and ③ statements because they are generally more convenient.

Spring Potential Energy: Whenever a spring is stretched or compressed, no energy is lost to atmosphere, the energy is retrieved when the spring regains its shape. Recall that we found the work done by internal conservative forces in a spring as $\Delta W_s = -\frac{1}{2} k x^2$ where x was elongation or compression from natural length. Thus potential energy corresponding to this conservative force:

$$\boxed{U_s = \frac{1}{2} k x^2}$$

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Finding Force Function From Potential Energy Function In Polar Form:

Recall that in cartesian form, $F_x = -\frac{\partial U}{\partial x}$, thus the component of force in any direction is simply negative of rate of change of potential energy per unit displacement in that direction. Thus if potential energy were expressed in polar coordinates $U(r,\theta)$, we can find the radial and tangential components of force by considering rates of change of potential energy in radial and tangential directions. Thus:

Radial Force: $\boxed{F_r = -\frac{\partial U}{\partial r}}$ and Tangential force $\boxed{F_\theta = \frac{-1}{r}\frac{\partial U}{\partial \theta}}$

(∵ keeping r constant, tangential displacement is $r d\theta$).

Centrifugal Potential Energy: While centrifugal force is not a real force, from rotating frames, it behaves as good as a real force. If a turn table is rotating with constant ω, the centrifugal force on a block at a distance r from centre is simply, $\vec{F_c} = m\omega^2 r\,\hat{r}$, thus centrifugal force is a central force in this frame and hence conservative. Now from previous section, for central forces, $F = -\frac{dU}{dr} \Rightarrow U = \int -F\,dr = \int -m\omega^2 r\,dr$ thus

$\boxed{U = -\frac{1}{2}m\omega^2 r^2}$ (We choose arbitrary constant as zero, which means, we assume zero PE at centre of the turn table).

→ Alternatively we can think of centrifugal force like force due to a point spring having a negative springs constant. For such a springs, $U = \frac{1}{2}kr^2$ where r is the instantaneous length. Here $k = -m\omega^2$.

Example: Potential Energy in a certain region of space is given as $U = \frac{K\cos\theta}{r^2}$ where K is some constant. Find the conservative force function in the polar form.

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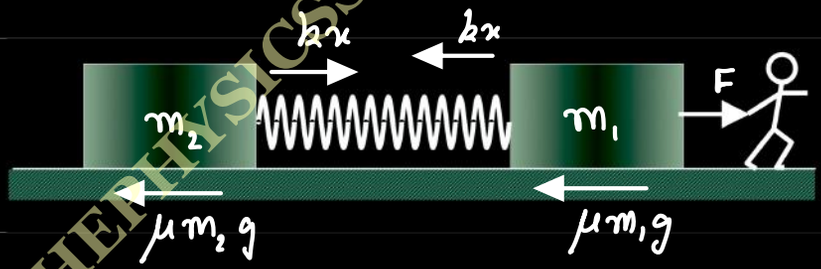
Solution: $F_r = -\frac{\partial U}{\partial r} = \frac{2K\cos\theta}{r^3}$, $F_\theta = -\frac{1}{r}\frac{\partial U}{\partial \theta} = \frac{K\sin\theta}{r^3}$

thus, $\vec{F} = \frac{K}{r^3}\left[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right]$ Ans.

EXAMPLE: IRODOV: 1.123:

Two bars of masses m_1 and m_2 connected by a non-deformed light spring rest on a horizontal plane. The coefficient of friction between the bars and the surface is equal to μ. What minimum constant force has to be applied in the horizontal direction to the bar of mass m_1 in order to shift the other bar?

Solution: In the limiting case, to nudge m_2, the extension required in the spring is such that $kx = \mu m_2 g$ ①



Now for extending the spring, m_1 must move to the right. Until F is more than resultant of kx and $\mu m_2 g$, the m_2 block will accelerate to the right. By the time forces equalize, it has already gained some KE and therefore continues to move rightward with some deceleration and finally will stop somewhere in accordance with WET (work energy theorem). The momentary stopping position of m_1 should be just sufficient to produce an extension required to nudge m_2. Now applying CHIME to m_1 + spring system:

$$Fx - \mu m_1 g x = \frac{1}{2}kx^2 \quad ② \quad (\because \text{KE is zero at momentary rest config's}).$$

Solving ① & ②: $F = \mu g\left[m_1 + m_2/2\right]$ Ans.

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EXAMPLE: IRODOV 1.124:

A chain of mass $m = 0.80$ kg and length $l = 1.5$ m rests on a rough-surfaced table so that one of its ends hangs over the edge. The chain starts sliding off the table all by itself provided the overhanging part equals $\eta = \frac{1}{3}$ of the chain length. What will be the total work performed by the friction forces acting on the chain by the moment it slides completely off the table?

Solution: Using pulling force method, in the limiting equilibrium, frictional force per unit length:

$$\lambda = \eta mg \times \frac{1}{l(1-\eta)}$$

Now suppose at some instant x length has slipped and further a dx length is slipping, in the process, work done by friction: $dW_f = -\lambda[(1-\eta)l - x]\,dx$, thus,

total work done by friction: $W_f = -\lambda \int_0^{(1-\eta)l} [(1-\eta)l - x]\,dx$

$$W_f = -\frac{1}{2}\eta(1-\eta)mgl \text{ Ans.} \qquad (= 1.3 \text{ Joule on putting values}).$$

→ Think: What is KE of the chain by the time it completely slips off the table.

Solution: Using CHIME, $W_f = \Delta KE + \Delta PE$, thus $\Delta KE = W_f - \Delta PE$.

Now PE can be found piecewise (CM are known piecewise), considering table surface as reference, $PE_i = 0 + (\eta mg)\left(-\eta\frac{l}{2}\right) = -\eta^2 mg\frac{l}{2}$

$PE_f = -mg\frac{l}{2}$ (finally whole chain is straight so no need for piecewise).

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Thus $DKE = -\frac{1}{2}\eta(1-\eta)mgl + mg\frac{l}{2} - \eta^2 mg\frac{l}{2} = \frac{mgl}{2}(1-\eta)$ Ans.

EXAMPLE: IRODOV : 1.131:

The potential energy of a particle in a certain field has the form $U = \frac{a}{r^2} - \frac{b}{r}$, where a and b are positive constants, r is the distance from the centre of the field. Find :

(a) the value of r_0 corresponding to the equilibrium position of the particle; examine whether this position is steady;

(b) the maximum magnitude of the attraction force; draw the plots $U(r)$ and $F_r(r)$ (the projection of the force on the radius vector r).

Solution: For equilibrium position, radial force must be zero $\therefore -\frac{\partial U}{\partial r} = 0$

$\Rightarrow \frac{-2a}{r^3} + \frac{b}{r^2} = 0 \Rightarrow r = 2a/b$. We will study about stability later on, however here we just use the fact that for stable equilibrium, potential energy should be at a local minimum.

Now: $\frac{d^2U}{dr^2} = \frac{6a}{r^4} - \frac{2b}{r^3} = \frac{1}{r^4}[6a - 2br] = \frac{1}{r^4}[6a - 4a] > 0$

Since second derivative of U is +ve $\therefore$ this must be a position of stable equilibrium. Now $\vec{F} = -\frac{dU}{dr}\hat{r} = \left(\frac{2a}{r^3} - \frac{b}{r^2}\right)\hat{r}$, this is attractive if the term in bracket is negative, i.e $\frac{2a}{r^3} - \frac{b}{r^2} < 0$

or $r > 2a/b$. For maximum force, $dF/dr = 0$

$\Rightarrow \frac{-6a}{r^4} + \frac{2b}{r^3} = 0 \Rightarrow r = 3a/b$ and corresponding value,

$F_{max} = -b^3/27a^3$ ($3a/b > 2a/b$ thus this is attractive force,

Also, $F = 0$ at $2a/b$ and also at ∞, thus force must be increasing and then decreasing back to zero $\therefore$ it must be a maximum.

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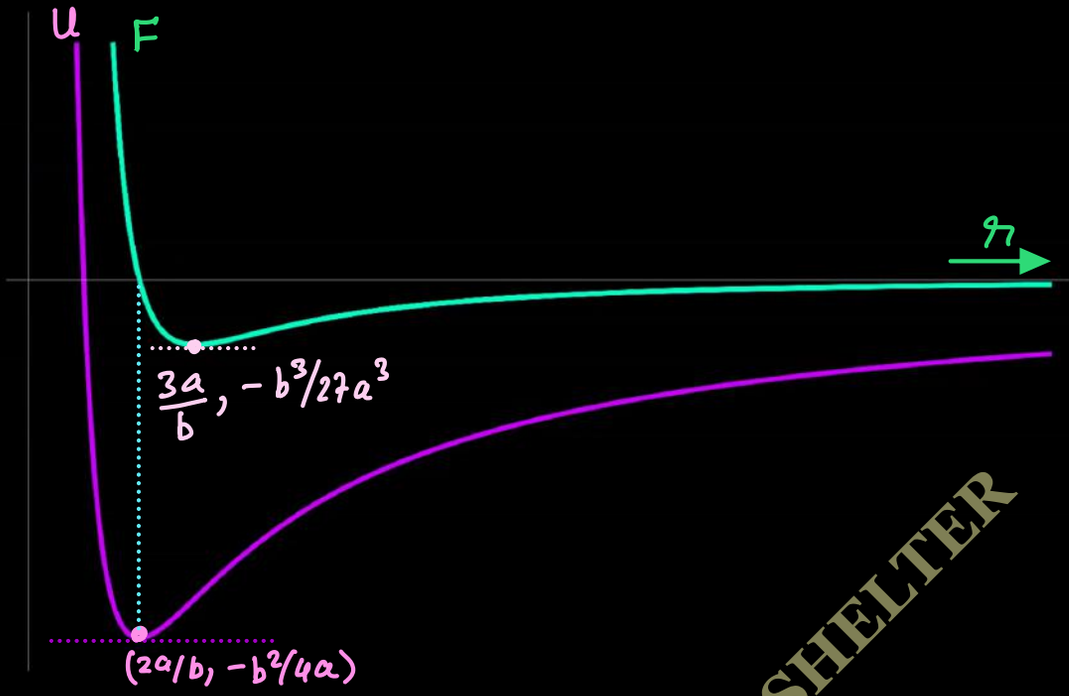
The plots of U and F are as shown:



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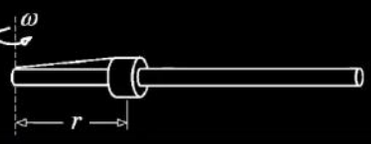
EXAMPLE: PATHFINDER: BUILD 20:

A horizontal frictionless thin rod wearing a sleeve of mass m is being rotated at a constant angular velocity ω about a stationary vertical axis through one of its ends. The sleeve is held stationary with respect to the rod at a distance $r = r_1$ from the axis with the help of a light cord as shown in the figure. At some instant of time, the cord is cut. Find work done by the external agency in maintaining the angular velocity of the rod constant until the sleeve slides to a distance $r = r_2$.



Solution: From the rotating frame, the only force that does work is the centrifugal force (coriolis force is always ⊥ to v_{rot} so does zero work). Since centrifugal force is a central force we can treat it like a conservative force from rotating frame. Now conserving mechanical energy from rotating frame:

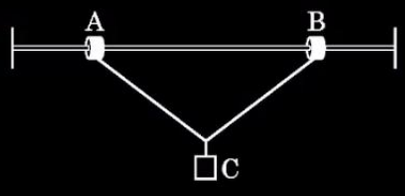
$$-\frac{1}{2} m r_1^2 \omega^2 = -\frac{1}{2} m r_2^2 \omega^2 + \frac{1}{2} m v_{rot}^2$$

$$\Rightarrow v_{rot} = \omega \sqrt{r_2^2 - r_1^2},$$ also from ground frame sleeve also has a tangential speed ωr_2 and initially it had a tangential speed ωr_1

Thus $W_{ext} = \Delta KE = \frac{1}{2} m \left[\left(\omega \sqrt{r_2^2 - r_1^2}\right)^2 + (\omega r_2)^2 - (\omega r_1)^2 \right]$

$$\Rightarrow \quad W_{ext} = \frac{1}{2} m\omega^2 \times 2 (r_2^2 - r_1^2) = m\omega^2 (r_2^2 - r_1^2) \text{ Ans.}$$

EXAMPLE: PATHFINDER: MCQ 18:



Two identical sleeves A and B each of mass m can slide without friction on a fixed horizontal rod. A load C of mass $2m$ is suspended from the mid-point of a light inextensible cord of length $l = 3\sqrt{3}$ m connecting the sleeves A and B. Initially the sleeves are at a separation l and the load is held motionless. What can you conclude for subsequent motion after the load is released? Acceleration of free fall is 10 m/s².

(a) Maximum speed of the sleeves is $\sqrt{10}$ m/s.
(b) Maximum speed acquired by the load C is $2\sqrt{5}$ m/s.
(c) The load descends a height 1.5 m to acquire maximum speed.
(d) Before the sleeves collide, the net force on the system consisting of the sleeves, the cord and the load becomes zero once.

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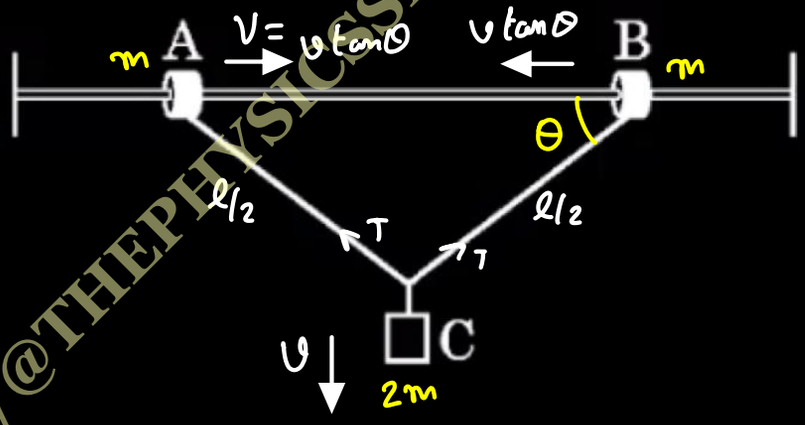


Solution: (a) Max speed of sleeves occurs when C reaches its lowest point because Tension is always doing positive work on sleeves. Also by velocity of separation concept, velocity of C at this instant must be zero. Thus by energy conservation:

$$2mg\frac{l}{2} = \frac{1}{2} m V^2 \times 2 \Rightarrow V_{max} = \sqrt{gl} = \sqrt{10 \times 3\sqrt{3}}$$ So (a) is wrong.

(b) Let max speed occur at an angular position θ as shown.
Thus by energy conservation:

$$\frac{1}{2} m (v\tan\theta)^2 \times 2 + \frac{1}{2} \times 2m v^2 = 2mg \frac{l}{2} \sin\theta$$

$$\Rightarrow \quad v^2 [1 + \tan^2\theta] = gl\sin\theta \Rightarrow v^2 = gl \sin\theta \cos^2\theta$$

For max v^2, we need to maximize $\sin\theta\cos^2\theta$, thus:

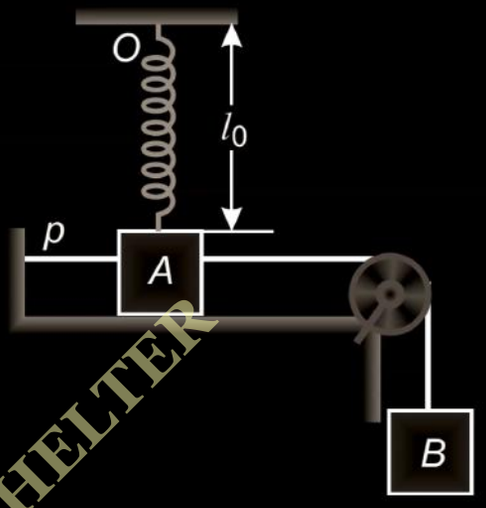
$$\cos^3\theta - 2\cos\theta \sin^2\theta = 0 \Rightarrow \tan\theta = 1/\sqrt{2},\ \sin\theta = 1/\sqrt{3},\ \cos\theta = \sqrt{\frac{2}{3}}$$

thus, $v_{max} = \sqrt{gl \sin\theta \cos^2\theta} = 2\sqrt{5}\ ms^{-1}$ thus (b) is correct.

(c) Descent of height $= l/2 \sin\theta = \frac{3\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = 1.5$ m (c) is correct

(d) When C acquires maximum speed, its acceleration is zero, thus, $\sum m_i a_{y_i} = 0$ therefore at this instant net force on the system must be zero.

EXAMPLE: IRODOV: 1.140: A small bar A resting on a smooth horizontal plane is attached by threads to point P and to a block B by means of a weightless pulley. A and B, both have mass m. Besides, the bar is also attached to point O by means of light non-deformed spring of length $l_0 = 50\,cm$ and stiffness $k = 5mg/l_0$. Once the thread connecting P and A is burnt, find the speed of A when it breaks off the plane.



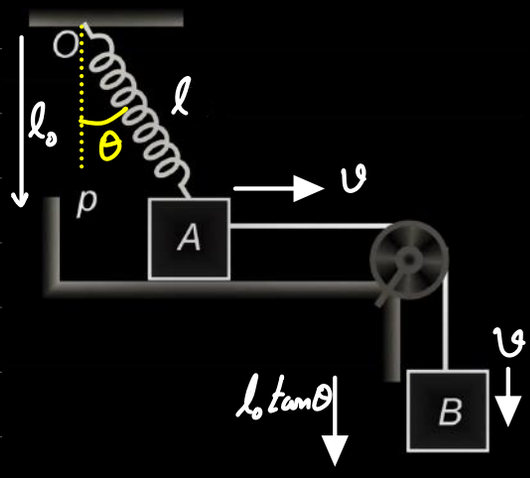
Solution: Let θ be the angle of the spring at which breaking off happens then, by force balance in vertical direction:

$$k l_0 (\sec\theta - 1)\cos\theta = mg$$

$$\Rightarrow 5(1-\cos\theta) = 1 \quad (\because k l_0 = 5mg).$$

$$\Rightarrow \cos\theta = \frac{4}{5},\ \theta = 37^\circ,\ l = \frac{5}{4} l_0,\ \Delta l = \frac{l_0}{4}$$

$$l_0 \tan\theta = \frac{3}{4} l_0$$



Now by energy conservation:

$$\frac{1}{2} k \left(\frac{l_0}{4}\right)^2 + \frac{1}{2} m v^2 \times 2 = mg \times \frac{3}{4} l_0$$

($\because$ loss in gravitational PE = Gain in KE + Spring PE).

$$\Rightarrow v = \sqrt{\frac{19 l_0 g}{32}} \text{ ans.}$$

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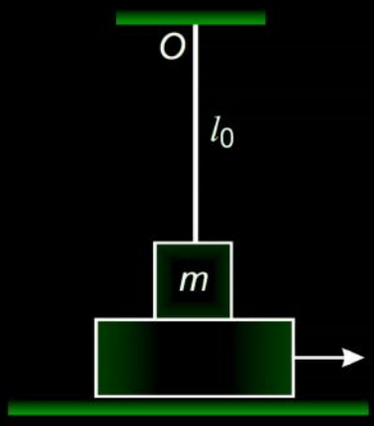
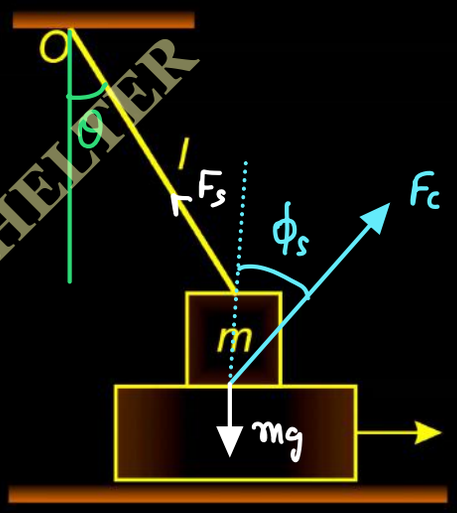
EXAMPLE : IRODOV : 1.141 :

A horizontal plane supports a plank with a bar of mass $m = 1.0$ kg placed on it and attached by a light elastic non-deformed cord of length $l_0 = 40$ cm to a point O (See Fig.). The coefficient of friction between the bar and the plank equal $\mu = 0.20$. The plank is slowly shifted to the right until the bar starts sliding over it. It occurs at the moment when the cord deviates from the vertical by an angle $\theta = 30°$. Find the work that has been performed by that moment by the friction force acting on the bar in the reference frame fixed to the plane.



Solution: At the verge of slipping, let contact force from plank be F_c. Then by Lami's theorem:

$$\frac{mg}{\sin(\theta+\phi_s)} = \frac{F_c}{\sin(180°-\theta)} = \frac{F_s}{\sin(180°-\phi_s)}$$



$$\Rightarrow F_s = \frac{mg\sin\phi_s}{\sin(\theta+\phi_s)}, \quad \text{Now work done}$$

by friction must be equal to energy stored in the spring i.e $\frac{1}{2} F_s \times \Delta l$, thus, $W_f = \frac{1}{2}\frac{mg\sin\phi_s}{\sin(\theta+\phi_s)} \cdot l_0(\sec\theta - 1)$

$$= \frac{mgl_0}{2}\left[\frac{\sin\phi_s}{\sin\theta\cos\phi_s + \cos\theta\sin\phi_s}\right] \cdot (\sec\theta - 1)$$

$$= \frac{mgl_0}{2}\left[\frac{\tan\phi_s}{\sin\theta + \cos\theta\tan\phi_s}\right][\sec\theta - 1] = \frac{mgl_0}{2}\left[\frac{\mu(\sec\theta - 1)}{\sin\theta + \mu\cos\theta}\right]$$

$$= \mu\frac{mgl_0}{2}\left[\frac{1-\cos\theta}{(\sin\theta+\mu\cos\theta)\cos\theta}\right] \quad \text{Ans.} \quad = 0.09 \text{ Joule.}$$

→ Note that spring energy is $\frac{1}{2}kx^2 = \frac{1}{2}(kx)\times x = \frac{1}{2}F_s x$.

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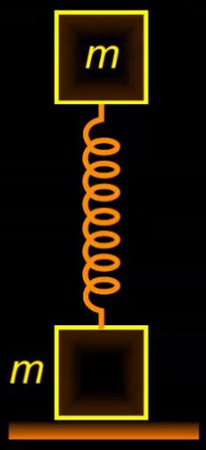
EXAMPLE: IRODOV: 1.153:

A system consists of two identical cubes, each of mass m, linked together by the compressed weightless spring of stiffness κ (See Fig.). The cubes are also connected by a thread which is burned through at a certain moment. Find:(a) at what values of Δl, the initial compression of the spring, the lower cube will bounce up after the thread has been burned through.



Solution: Let x be the required compression and y be the max height attained by upper block in limiting case. Then:

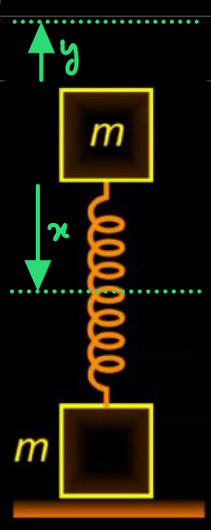
$$ky = mg \Rightarrow y = mg/k \quad (1)$$

Conserving energy between levels A & B:

$$\frac{1}{2}kx^2 - \frac{1}{2}ky^2 = mg(x+y) \quad (2)$$

Now from (1) & (2):

$x = 3mg/k$ Ans. Thus required $\Delta l > 3mg/k$ Ans.



EXAMPLE:

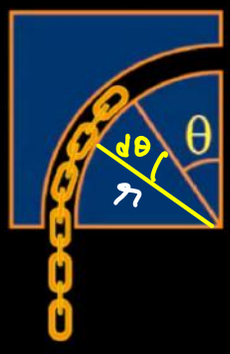
The flexible chain of length $\pi r/2$ and mass per unit length λ is released from rest with $\theta = 0°$ in the smooth circular channel and falls through the hole in the supporting surface. Determine the speed of the chain as the last link leaves the slot.

(A) $\sqrt{gr\left(\frac{\pi}{2}+\frac{2}{\pi}\right)}$ (B) $\sqrt{gr\left(\frac{\pi}{2}+\frac{4}{\pi}\right)}$

(C) $\sqrt{gr\left(\frac{\pi}{4}+\frac{4}{\pi}\right)}$ (D) $\sqrt{gr\left(\frac{\pi}{2}+\frac{1}{\pi}\right)}$

Solution: Loss in PE must equal gain in KE. For initial PE, we will have to integrate. Consider an element of angular width $d\theta$. Elemental PE:

$$dU_i = \frac{m}{\pi/2} \cdot d\theta \cdot g \cdot r\sin\theta \quad (\because \text{reference level at centre of arc}).$$



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$$\Rightarrow U_i = \frac{2m}{\pi} g r \int_0^{\pi/2} \cos\theta \, d\theta = 2mgr/\pi.$$

Finally CM is at a depth, $\pi r/4$ below the centre of arc. Thus,

$U_f = -mg\frac{\pi r}{4}$. Now by COME: $mgr\left[\frac{2}{\pi} + \frac{\pi}{4}\right] = \frac{1}{2}mv^2$

$$\Rightarrow v = \sqrt{gr\left(\frac{4}{\pi} + \frac{\pi}{2}\right)}$$ option B also.

Finding Potential Energy Function from Force Function:

Recall that for finding force function from potential energy function, we had to do partial differentiation, thus to find potential energy function from force function, we will have to do partial integration. Just as while doing usual integration, we have to add an arbitrary constant, while doing partial integration, we have to add an arbitrary function of other variables. For instance if $H(x,y,z)$ is some function of three variables and $\frac{\partial H}{\partial x} = \cos x$ then we can infer that H must have a form $H = \sin(x) + f(y,z)$ where $f(y,z)$ is an arbitrary function of y and z. This can be readily verified by once again taking partial derivative of H wrt x and we see that $\partial H/\partial x = 0$ and therefore we get $\partial H/\partial x = \cos(x)$. If $\vec{F}(x,y,z)$ is a conservative force, we need to find a function $U(x,y,z)$ such that:

$$\frac{\partial U}{\partial x} = -F_x \;, \frac{\partial U}{\partial y} = -F_y \text{ and } \frac{\partial U}{\partial z} = -F_z.$$

U can be found by suitably using partial integration to simultaneously satisfy all the three conditions.

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EXAMPLE: A two dimensional force function in space is given as $\vec{F} = (2x+y)\hat{i} + x\hat{j}$ (a) Is it conservative (b) If yes find the potential energy function assuming zero PE at the origin.

Sol: (a) $\frac{\partial F_x}{\partial y} = 1$ $\frac{\partial F_y}{\partial x} = 1$ thus $\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}$ $\therefore$ it is conservative

(b) $F_x = \frac{-\partial U}{\partial x} \Rightarrow \frac{\partial U}{\partial x} = -(2x+y)$ (1) and $\frac{\partial U}{\partial y} = -x$ (2)

From first condition, $U = -[x^2 + xy] + f(y)$ now using this in second condition:

$$\frac{\partial U}{\partial y} = -x + \frac{df}{dy} = -x \Rightarrow \frac{df}{dy} = 0 \Rightarrow f = C$$

(Note: here we used ordinary derivative of f because it is single valued).

Thus our required PE function becomes $U = -[x^2+xy] + C$, however as per our chosen reference, $U(0,0) = 0 \Rightarrow C = 0$ thus: $U = -(x^2+xy)$ Ans.

→ Caution: Do not integrate the two equations separately and add because that would lead to duplication of terms containing x and y simultaneously.

→ Always integrate first equation and use the result in the second equation to find unknown parts of potential energy function.

EXAMPLE: A two dimensional force field is given as $\vec{F} = y\hat{i} - x\hat{j}$, try to find the potential energy function.

Solution: $\partial F_x/\partial y = 1$ $\partial F_y/\partial x = -1$ so it is a non conservative force and therefore potential energy function is not defined. (You can try to use the algorithm for finding U but you will run in to contradiction).

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EXAMPLE: A 3D force function is given as $\vec{F} = -[2x\hat{i} + z\hat{j} + (y+2z)\hat{k}]$ (a) is the force conservative (b) If yes find PE function assuming zero PE at the origin.

Solution: For conservative forces curl is zero i.e $\vec{\nabla} \times \vec{F} = 0$ (identically), Now:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x & z & (y+2z) \end{vmatrix} = \hat{i}[1-1] - \hat{j}[0-0] + \hat{k}[0-0] = 0 \quad \text{(identically).}$$

Thus force is conservative. (b) Now $\frac{\partial u}{\partial x} = 2x \Rightarrow u = x^2 + f(y,z)$

using it in second partial derivative condition, $\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} = z$

$\Rightarrow f = yz + g(z)$, thus $u = x^2 + yz + g(z)$ now using this in 3rd partial derivative condition, $\frac{\partial u}{\partial z} = y + \frac{dg}{dz} = y + 2z \Rightarrow \frac{dg}{dz} = 2z$

$\Rightarrow g = z^2 + c$, thus our function becomes $u = x^2 + yz + z^2 + c$

Now since $U(0,0,0) = 0 \Rightarrow c = 0$ thus $u = x^2 + yz + z^2$ Ans.

FINDING WORK DONE BY NON CONSERVATIVE FORCES: For conservative forces, very often easiest way to find work done is by finding potential energy function and then simply using $W_c = -\Delta U$, there we need not worry about exact path. However with non-conservative forces, work will depend on path and we must specify the path in going from one point to another. In a 2D plane let a non conservative force be $\vec{F} = F_x\hat{i} + F_y\hat{j}$ where F_x and F_y themselves may be functions of x and y and let an elemental displacement be $d\vec{s} = dx\hat{i} + dy\hat{j}$, then the elemental work is given as $dW = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy$. Now, finding work

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by integrating this would involve a mixed integral in variables x and y, however if we know the path equation $y = f(x)$ then the whole integral can be converted to a single variable using $dy = \frac{df}{dx}\,dx$ (or alternatively if $x = g(y)$ then $dx = \frac{dg}{dy}\,dy$ and whole integral can be transformed to variable y).

EXAMPLE: A certain force function is given as $\vec{F} = \alpha y^2 \hat{i} + \beta x^2 \hat{j}$. Find the work done in going from $(0,0)$ to (a, a^2) along path $y = x^2$.

Solution: $\vec{F} \cdot d\vec{s} = \alpha y^2 dx + \beta x^2 dy \quad (\because d\vec{s} = dx\,\hat{i} + dy\,\hat{j})$.

Now $y = x^2 \Rightarrow dy = 2x\,dx$ thus transforming everything to x,

$$dW = \alpha x^4 dx + \beta x^2 \cdot 2x\,dx \Rightarrow W = \int_0^a (\alpha x^4 + 2\beta x^3)\,dx$$

$$= \frac{\alpha a^5}{5} + \frac{\beta a^4}{2} \text{ Ans.}$$

Think: For same force what is the work done along line $y = ax$

Solution: $y = ax \Rightarrow dy = a\,dx$ thus transforming our elemental work:

$$dW = \alpha (ax)^2 dx + \beta x^2 (a\,dx) = a^2 \alpha x^2 dx + a\beta x^2 dx$$

Thus $W = \int_0^a a x^2 [a\alpha + \beta]\,dx = \frac{a^4}{3}[a\alpha + \beta]$ Ans.

→ Clearly the work done along two paths is different.

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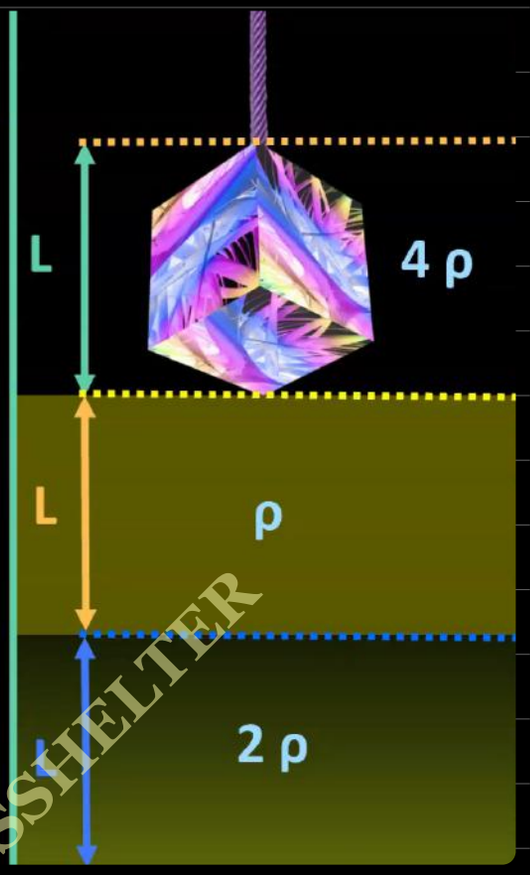
THE DROWNING CUBE CHALLENGE

Challenge: A cube having body diagonal of length L and density 4ρ (Mass 4M) is being slowly lowered in to a very wide pool containing two liquids as shown. Calculate the work done by the tension in process from the moment the bottom vertex of the cube is touching surface of the top liquid to the moment when the bottom vertex just touches the bottom of lower liquid.



PHYSICS CHALLENGES With Ambarish

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Solution: Work done must be equal to change in potential energy of the system (if movements are slow enough to ignore any KE changes/ splashing etc). Here cube is losing PE while displaced molecules of liquid with 2ρ rise to its top surface and similarly molecules displaced of liquid of density ρ rise to form its top layer. This gives us the following visualization and solution:

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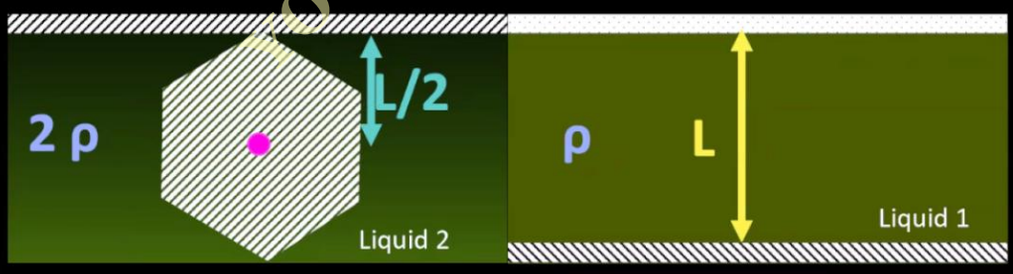
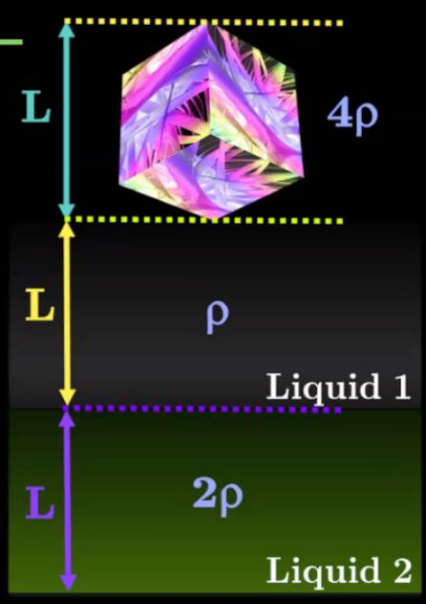
Work Done= Change in Mechanical Energy of the system





$$\Delta W = \Delta PE_2 + \Delta PE_1 + \Delta PE_{cube}$$

$$= 2Mg \times \frac{L}{2} + Mg \times L - 4Mg \times 2L$$

$$\boxed{\Delta W = -6MgL \quad \text{Ans.}}$$

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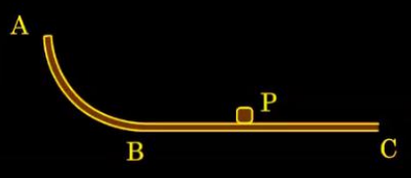
Example: Pathfinder: MCQ13:



A frictionless rod ABC and a particle P of mass m are lying on a frictionless horizontal floor as shown in top view of the situation. The portion AB of the rod is bent to form a quarter of a circle. The rod is suddenly made to move with a constant velocity v towards the right along portion BC. What is the work done by the rod on the particle till it leaves the rod?

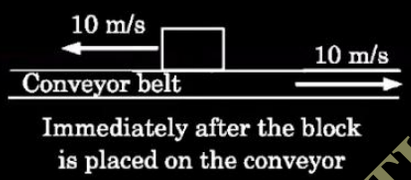
(a) $0.5mv^2$ (b) mv^2

(c) $2mv^2$ (d) None of these.

Solution: From rod frame, the particle starts moving with a speed v and exits from A with same speed, but from ground frame, it also has a horizontal component due to speed of the rod itself, thus work done on particle = DKE: $\frac{1}{2}m(v^2+v^2) = mv^2$ (option b).

Example: Pathfinder: MCQ 19:

A box of mass 1.0 kg moving horizontally with velocity a 10 m/s towards the left is gently placed on a conveyor belt that is moving horizontally with a velocity 10 m/s towards the right. Coefficient of kinetic friction between the belt and the block is 0.5. Some physical quantities and their possible value in SI units are mentioned in the given table. Suggest suitable matches.



Immediately after the block is placed on the conveyor

Modulus in SI	Physical Quantity
(a) zero	(p) Work of friction on the block with respect to the ground
(b) 20	(q) Work of friction on the block with respect to the belt
(c) 40	(r) Work of friction on the belt with respect to the ground
(d) 200	(s) Distance slid by the block on the belt with respect to the belt
	(t) Change in momentum of the block with respect to the ground

Solution: (b) Since initial KE and final KE will be same (even though velocity is reversed), the net work done by friction in ground frame must be zero $\therefore$ p → a thus.

(d) With respect to belt, final KE is zero while initially it is $\frac{1}{2}\times1\times(10+10)^2$ = 200 J thus from conveyor frame work done by friction is

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DKE = −200 J thus q → d ans.

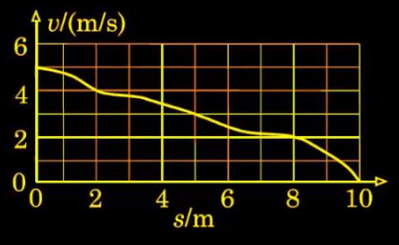
(iii) The work of kinetic pair is frame independent, thus work done by kinetic friction pair in ground frame must be same as that in conveyor frame i.e −200 J. Thus $-200 = W_{belt} + W_{block}$
$\Rightarrow W_{belt} = -200$ thus p → d ans.

(s) Deceleration = $\mu g = 5\ ms^{-2}$, thus distance slid wrt conveyor is found using, $0^2 - 20^2 = 2 \times (-5) \times s \Rightarrow s = 40$ m thus s → c ans

(t) Change in momentum = $m\Delta v = 1 \times (10 - (-10)) = 20\ kg\ ms^{-1}$ thus t → b ans

EXAMPLE : PATHFINDER: BUILD:4:

On a horizontal surface of non-uniform texture, friction varies from point to point. On this surface a small disc projected with speed 5 m/s covers 10 m along a straight path before it stops. Variation in speed v of the disc with distance s covered is shown in the figure. If the disc is projected with speed 4 m/s from the same initial point in the same direction, how much total distance will it cover?



Solution: Irrespective of launching speed, so long as friction is kinetic, loss in KE is same for the same sliding distance. Now $KE \propto v^2$, thus we need to locate a point on graph where loss in value of v^2 is 16. For first disc the initial value of v^2 is 25, thus for a loss of 16 units in v^2, the v^2 must be reduced to 9 or v must reduce to 3. This corresponds to s = 5m. Thus our required answer: s = 5m ans.

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